

From Local to Global: Revisiting Structured Pruning Paradigms for Large Language Models

从局部到全局：重新审视大型语言模型的结构化剪枝范式

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Abstract

摘要

Structured pruning is a practical approach to deploying large language models (LLMs) efficiently, as it yields compact, hardware-friendly architectures. However, the dominant local paradigm is task-agnostic: by optimizing layer-wise reconstruction rather than task objectives, it tends to preserve perplexity or generic zero-shot behavior but fails to capitalize on modest task-specific calibration signals, often yielding limited downstream gains. We revisit global structured pruning and present GISP, Global Iterative Structured Pruning, a post-training method that removes attention heads and MLP channels using first-order, loss-based importance scores aggregated at the structure level with block-wise normalization. Built on this global importance metric, GISP adopts an iterative schedule, rather than one-shot pruning, stabilizes accuracy at higher sparsity, and mitigates perplexity collapse without requiring intermediate fine-tuning. Importantly, the iterative pruning forms nested subnetworks that support a “prune-once, deploy-many” workflow. Furthermore, GISP defines structural importance directly with respect to a target loss, making it easy to adapt pruning to task-specific objectives. In this work, we use perplexity for language modeling and a margin-based objective for decision-style tasks. Extensive experiments show that across Llama2-7B/13B, Llama3-8B, and Mistral-0.3-7B, GISP consistently lowers WikiText-2 perplexity and improves downstream accuracy, with especially strong gains at 40–50% sparsity; on DeepSeek-R1-Distill-Llama-3-8B and Qwen3-8B with GSM8K, task-aligned calibration substantially boosts exact-match accuracy. The implementation is available at the official GISP repository.

结构化剪枝是一种实用的方法，用于高效部署大型语言模型（LLMs），因为它能够生成紧凑且适合硬件的架构。然而，主导的局部范式是任务无关的：通过优化逐层重建而不是任务目标，它倾向于保留困惑度或通用的零样本行为，但无法充分利用适度的任务特定校准信号，通常导致下游收益有限。我们重新审视全局结构化剪枝，并提出了 GISP（全局迭代结构化剪枝），这是一种后训练方法，通过在结构层面上使用块状归一化的基于一阶损失的重要分数，来移除注意力头和 MLP 通道。基于这一全局重要性度量，GISP

采用迭代计划，而不是一次性剪枝，从而在更高稀疏度下稳定准确率，并在不需要中间微调的情况下缓解困惑度崩溃。重要的是，迭代剪枝形成了嵌套的子网络，支持“一次剪枝，多次部署”的工作流程。此外，GISP 直接根据目标损失定义结构重要性，使其易于适应任务特定的目标。在本工作中，我们使用困惑度进行语言建模，并使用基于边距的目标处理决策型任务。大量实验表明，在 Llama2-7B/13B、Llama3-8B 和 Mistral-0.3-7B 上，GISP 始终降低 WikiText-2 的困惑度并提高下游准确率，特别是在 40–50% 稀疏度时有显著提升；在 DeepSeek-R1-Distill-Llama-3-8B 和 Qwen3-8B 与 GSM8K 任务中，任务对齐的校准显著提升了精确匹配准确率。该实现可在官方 GISP 仓库中获取。

1 Introduction

1 引言

Pruning (Ma et al., 2023; Frantar and Alistarh, 2023; Sun et al., 2024; Kim et al., 2024; An et al., 2023) is a fundamental technique for compress-

剪枝 (Ma 等, 2023; Frantar 和 Alistarh, 2023; Sun 等, 2024; Kim 等, 2024; An 等, 2023) 是压缩-

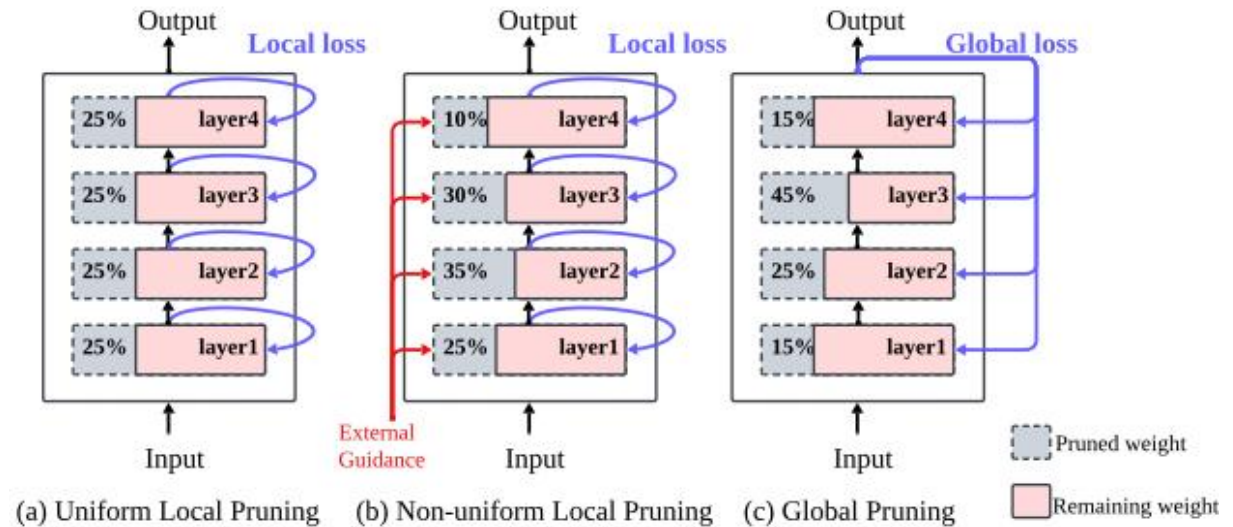


Figure 1: images/fd46af9963652b857a03449c2171a0b91acba0983a77e8a97af63d9e10b651b3.jpg

Figure 1: Comparison between (a) local pruning, which uses layer-wise reconstruction loss as the importance criterion, (b) non-uniform variants of local pruning, and (c) global pruning, which directly considers the impact of weight pruning on the final model loss.

图 1: (a) 局部剪枝，其使用逐层重建损失作为重要性指标，(b) 非均匀局部剪枝的变体，以及 (c) 全局剪枝，其直接考虑权重剪枝对最终模型损失的影响。

ing neural networks by removing redundant parameters while preserving accuracy. Broadly, existing approaches fall into two categories: unstructured pruning, which removes element-wise weights without shrinking the model architecture, and structured pruning, which eliminates entire groups of weights (e.g., channels, attention heads, layers). It is well established that unstructured pruning can achieve higher sparsity levels but typically requires specialized sparse computation kernels to realize runtime speedups, whereas structured pruning inherently produces compact, hardware-friendly architectures and is therefore preferred for practical deployment. With the rapid emergence of Large Language Models (LLMs) (Touvron et al., 2023a,b; OpenAI et al., 2024; Chiang et al., 2023; Workshop et al., 2023; Grattafiori et al., 2024) containing billions of

parameters, pruning has become critical to improve inference efficiency and to enable deployment on resource-constrained devices.

通过移除冗余参数同时保持准确性来压缩神经网络。总体而言，现有方法大致分为两类：无结构剪枝 (unstructured pruning)，它在不缩小模型结构的情况下移除逐元素权重；以及有结构剪枝 (structured pruning)，它消除整个权重组（例如通道、注意力头、层）。众所周知，无结构剪枝可以实现更高的稀疏性水平，但通常需要专门的稀疏计算内核来实现运行时加速，而有结构剪枝则会自然地产生紧凑且适合硬件的架构，因此更适用于实际部署。随着包含数十亿参数的大型语言模型 (LLMs) 的迅速出现 (Touvron 等, 2023a,b; OpenAI 等, 2024; Chiang 等, 2023; Workshop 等, 2023; Grattafiori 等, 2024)，剪枝变得至关重要，以提高推理效率并实现在资源受限设备上的部署。

The dominant paradigm for pruning LLMs is local pruning, exemplified by methods like SparseGPT (Frantar and Alistarh, 2023) and Wanda (Sun et al., 2024). These methods gained significant attention due to their simplicity and efficiency, breaking down the model-wide optimization into layer-wise sub-problems (Fig.1(a)). This decomposition allows them to prune each layer

剪枝大型语言模型 (LLMs) 的主流范式是局部剪枝，例如 SparseGPT (Frantar 和 Alistarh, 2023) 和 Wanda (Sun 等, 2024) 等方法。这些方法由于其简单性和高效性而获得了广泛关注，它们将模型整体优化分解为逐层的子问题（见图 1(a)）。这种分解使它们能够逐层进行剪枝

gradually, typically by minimizing a layer-wise reconstruction with calibration data, offering a post-training solution. Furthermore, to mitigate the rigidity of uniform sparsity, recent work explores non-uniform local pruning (Fig. 1(b)), which adjusts layer-wise ratios based on estimated importance. Methods such as OWL (Yin et al., 2025), FLAP (An et al., 2023), and DarwinLM (Tang et al., 2025) leverage activation statistics or evolutionary search to assign non-uniform sparsity. While these approaches improve accuracy, they remain rooted in layer-wise reconstruction and introduce notable algorithmic complexity and overhead.

逐步地，通常通过使用校准数据最小化逐层重建，提供了一种后训练解决方案。此外，为了缓解均匀稀疏性的刚性，近期的研究探索了非均匀局部剪枝（图 1(b)），该方法根据估计的重要性调整各层的稀疏率。例如，OWL (Yin 等, 2025)、FLAP (An 等, 2023) 和 DarwinLM (Tang 等, 2025) 等方法利用激活统计或进化搜索来分配非均匀稀疏性。虽然这些方法提高了准确性，但它们仍然基于逐层重建，并引入了显著的算法复杂性和开销。

While local structured pruning is efficient and preserves broad behavior (often reflected in perplexity or generic zero-shot scores), its objective is task-agnostic. When modest task-informed calibration is available, local methods rarely capitalize on it, yielding limited downstream gains. This gap calls for a loss-aligned alternative defined at the model level, instead of a local proxy. We therefore revisit global pruning (Fig.1(c)) for LLMs and develop GISP—Global Iterative Structured Pruning. Unlike local approaches that optimize layer reconstructions, global pruning defines importance with respect to a model-level loss, naturally inducing non-uniform sparsity without extra heuristics. Operationally, GISP aggregates first-order, loss-based importance at the structure level (attention heads and MLP channels) with block-wise normalization, and we study it in a post-training setting (no fine-tuning between steps) to match practical constraints.

虽然局部结构化剪枝在效率和保持广泛行为方面表现良好（通常体现在困惑度或通用零样本得分上），但其目标与任务无关。当有适度的任务相关信息校准时，局部方法很少利用这些信息，导致下游收益有限。这一差距促使我们提出一种与损失对齐的替代方案，该方案在模型层面定义，而非局部代理。因此，我们重新审视了大语言模型 (LLMs) 的全局剪枝（图 1(c)），并开发了 GISP——全局迭代结构化剪枝。与优化层重建的局部方法不同，全局剪枝基于模型层面的损失定义重要性，自然地诱导非均匀稀疏性，而无需额外的启发式方法。在操作上，GISP 通过块级归一化，在结构层面（注意力头和 MLP 通道）聚合基于损失的一阶重要性，并在后训练设置下研究它（步骤之间不进行微调），以匹配实际约束。

Building on this formulation, we first validate that making global pruning iterative fundamentally changes its behavior. A gradual, ratio-scheduled process turns the otherwise unstable one-shot global pruning into a robust procedure that preserves model quality even at high sparsity. Furthermore, the same iterative trajectory also reveals a nested structure across sparsity levels, showing that iterative global pruning can serve as a single, continuous optimization rather than

a series of independent runs, thereby enabling a “prune-once, deploy-many” workflow. Finally, because importance is defined by a model-level loss, GISP can directly integrate task objectives, bridging the gap between generic compression and task-aware optimization; this property consistently yields stronger downstream accuracy across models and pruning ratios.

基于这一表述，我们首先验证了进行全局剪枝的迭代过程从根本上改变了其行为。一个渐进的、基于比率的调度过程，将原本不稳定的单次全局剪枝转变为一种稳健的程序，即使在高稀疏度下也能保持模型质量。此外，相同的迭代轨迹还揭示了不同稀疏度层级之间的嵌套结构，表明迭代式全局剪枝可以作为一种连续的优化过程，而不是一系列独立的运行，从而实现“一次剪枝，多次部署”的工作流程。最后，由于重要性是由模型级别的损失定义的，GISP 可以直接集成任务目标，弥合通用压缩与任务感知优化之间的差距；这一特性在不同模型和剪枝比例下始终能够带来更强的下游准确性。

We summarize our contributions as follows:

我们总结我们的贡献如下：

- We present GISP, a simple and effective global iterative structured pruning framework for LLMs that operates post-training and stabilizes performance at high sparsity.
- We demonstrate that iterative global pruning follows a smooth, nested trajectory of sub-networks, enabling a “prune-once, deploy-many” workflow with a competitive amortized time cost per usable model compared to local pruning baselines.
- We examine the task-specific property of GISP and instantiate task-specific global importance via multiple objectives. Extensive experiments demonstrate consistent downstream gains across models and pruning ratio levels.
- 我们提出 GISS，一个简单且有效的全局迭代结构化剪枝框架，用于 LLMs，在训练后运行，并在高稀疏度下稳定性能。
- 我们证明了迭代全局剪枝遵循一个平滑的、嵌套的子网络轨迹，使得可以实现“一次剪枝，多次部署”的工作流程，其每可用模型的平均时间成本与本地剪枝基线相比具有竞争力。
- 我们研究了 GISS 的任务特定属性，并通过多个目标实例化任务特定的全局重要性。大量实验表明，在不同模型和剪枝比例级别下，下游性能均有持续提升。

2 Preliminary: Local vs Global Pruning

2 预备知识：局部剪枝与全局剪枝

2.1 Local Pruning and Non-uniform Variant

2.1 局部剪枝与非均匀变体

Rationale of local pruning. Given a pre-trained model with weights as θ and a set of calibration dataset $D = \{(x_i, y_i)\}_{i=1}^N$ with N samples, the structure pruning in LLMs with L transformer (Vaswani et al., 2023) layers can be interpreted as finding optimal $\hat{\theta}$ under desired sparsity constraints by removing sets of coupled structures W_l from $G = (\{W_{l, \text{Attn}}\}_{l=1}^L, \{W_{l, \text{MLP}}\}_{l=1}^L)$ with minimal error on a pre-defined objective function.

局部剪枝的原理。给定一个预训练模型，其权重为 θ ，以及一个包含 N 个样本的校准数据集 $D =$

$\{(x_i, y_i)\}_{i=1}^N$ ，在具有 L 层 Transformer (Vaswani 等人, 2023) 的大型语言模型中，结构剪枝可以被解释为：通过从 $G = (\{W_{l,\text{Attn}}\}_{l=1}^L, \{W_{l,\text{MLP}}\}_{l=1}^L)$ 中移除一组耦合结构 W_l ，在预定义的目标函数上以最小的误差，寻找满足所需稀疏性约束的最优 $\hat{\theta}$ 。

As introduced by the pioneering work OBS (Hassibi and Stork, 1992) and layer-wise OBS (Dong et al., 2017), local pruning defines the objective function by breaking down the problem of full model compression into sub-problems for each layer. It constructs a local loss to measure the L_2 error between the outputs of the unpruned and pruned layers, which can be formulated as:

如由开创性工作 OBS (Hassibi 和 Stork, 1992) 以及逐层 OBS (Dong 等, 2017) 所介绍，局部剪枝通过将完整模型压缩问题分解为每一层的子问题来定义目标函数。它构建了一个局部损失函数，以衡量未剪枝层和剪枝层输出之间的 L_2 误差，其可以表示为：

$$\min_{M_\ell, \widehat{W}_\ell} \|W_\ell X_\ell - (M_\ell \odot \widehat{W}_\ell) X_\ell\|_2^2, \quad (1)$$

where W_ℓ is the original weight of layer ℓ , X_ℓ is the input to layer ℓ , M_ℓ is the binary mask indicating which weights to keep, and $\widehat{W}_\ell$ denotes the postpruning weights, which may be kept fixed as $\widehat{W}_\ell$ or re-optimized given M_ℓ .

其中 W_ℓ 是第 ℓ 层的原始权重， X_ℓ 是输入到第 ℓ 层的输入， M_ℓ 是二进制掩码，用于指示哪些权重需要保留， $\widehat{W}_\ell$ 表示剪枝后的权重，这些权重可以保持为 W_ℓ 不变，或者根据 M_ℓ 进行重新优化。

Among local structured pruning methods, SparseGPT (Frantar and Alistarh, 2023) formulates pruning as a sparse regression problem solved via an approximate Hessian inversion. ZipLM (Kurtic et al., 2023) extends the OBS formulation to structured pruning and performs inference-aware search

在局部结构化剪枝方法中，SparseGPT (Frantar and Alistarh, 2023) 将剪枝表述为一个稀疏回归问题，并通过近似 Hessian 矩阵求逆来求解。ZipLM (Kurtic et al., 2023) 将 OBS 表述扩展到结构化剪枝，并执行面向推理的搜索

over structures. Wanda (Sun et al., 2024) simplifies SparseGPT’s importance to weight–activation products, achieving similar accuracy with higher efficiency. LLM-Pruner (Ma et al., 2023) further prunes entire attention heads and MLP channels using gradient information to capture interstructure dependencies. GBLM (Das et al., 2024) augments Wanda’s scores with gradient magnitude, while Wanda++ (Yang et al., 2025b) adds block-level gradients from activation reconstruction to improve layer-wise pruning; Pruner-Zero (Dong et al., 2024) instead uses genetic programming to discover gradient-based layer-local importance metrics from weight/activation/gradient signals. Finally, several works explore layer-wise pruning (Kim et al., 2024; Men et al., 2024), such as ShortGPT (Men et al., 2024), which leverages layer-wise activation similarity.

在结构上。Wanda (Sun 等, 2024) 将 SparseGPT 的重要性简化为权重-激活乘积，实现了与之相近的精度，但效率更高。LLM-Pruner (Ma 等, 2023) 进一步利用梯度信息剪除整个注意力头和 MLP 通道，以捕捉结构间的依赖关系。GBLM (Das 等, 2024) 通过梯度幅度增强 Wanda 的评分，而 Wanda++ (Yang 等, 2025b) 则通过激活重建引入块级梯度，以改进逐层剪枝；Pruner-Zero (Dong 等, 2024) 则使用遗传编程，从权重/激活/梯度信号中发现基于梯度的层内重要性指标。最后，一些研究探索了逐层剪枝 (Kim 等, 2024; Men 等, 2024)，例如 ShortGPT (Men 等, 2024)，其利用了逐层激活相似性。

Non-uniform variants of local pruning. To overcome the limitation of uniform sparsity in layer-wise pruning, several works introduce non-uniform local pruning (Fig. 1(b)) that adjusts pruning ratios across layers based on estimated importance. These methods extend the layer-wise reconstruction paradigm by incorporating inter-layer sensitivity through diverse heuristics: FLAP (An et al., 2023) exploits activation variability to assign flexible sparsity, OWL (Yin et al., 2025) reweights layers according to outlier statistics in activations, and DarwinLM (Tang et al., 2025)

performs a training-aware evolutionary search to identify optimal sparsity configurations.

非均匀的局部剪枝。为了克服逐层剪枝中均匀稀疏性的限制，一些工作引入了非均匀局部剪枝（图 1(b)），该方法根据估计的重要性在不同层之间调整剪枝比例。这些方法通过多样化的启发式方法引入层间敏感性，扩展了逐层重建范式：FLAP（An 等人，2023）利用激活的变异性来分配灵活的稀疏性，OWL（Yin 等人，2025）根据激活中的异常值统计重新加权层，而 DarwinLM（Tang 等人，2025）通过训练感知的进化搜索来识别最优的稀疏性配置。

2.2 Global Pruning

2.2 全局剪枝

Global pruning aims to find a global sparsity mask M and possibly updated weights $\widehat{W}$ to minimize the global loss between the final outputs of the uncompressed and compressed model. Hence, the learning objective can be formulated as:

全局剪枝旨在找到一个全局稀疏性掩码 M 和可能更新的权重 $\widehat{W}$ ，以最小化未压缩模型和压缩模型最终输出之间的全局损失。因此，学习目标可以表示为：

$$\min_{M, \widehat{W}} \Delta \mathcal{L} \left(f(X; M \odot \widehat{W}), f(X; W) \right), \quad (2)$$

where f is the forward function, X denotes the inputs, W is the original (pre-trained) weight, M is the binary mask indicating which weights remain. Following the idea from OBD (LeCun et al., 1989) of conducting the Taylor series towards loss distance on parameter perturbation caused by pruning, we have element-wise importance given by

其中 f 是前向函数， X 表示输入， W 是原始（预训练）权重， M 是二进制掩码，表示哪些权重保留下来。根据 OBD (LeCun 等人，1989) 的思想，通过剪枝引起的参数扰动在损失距离上进行泰勒展开，我们得到逐元素的重要性如下：

$$I_{W_i^j} = |\Delta \mathcal{L}(D)| = \left| \mathcal{L}(D; \theta_{W_i^j}) - \mathcal{L}(D; \theta_{W_i^j=0}) \right| = \left| \frac{\partial \mathcal{L}(D)}{\partial W_i^j} W_i^j - \frac{1}{2} W_i^j H_{jj} W_i^j + \mathcal{O}(\|W_i^j\|^3) \right|, \quad (3)$$

where $I_{W_i^j}$ marks the j -th estimated importance of element in θ , H_{jj} is diagonal of the hessian matrix. Global pruning has been extensively studied in smaller networks such as CNNs, Vision Transformers, and compact language models (Molchanov et al., 2016; Yang et al., 2023; Diao et al., 2023), consistently outperforming local approaches (Blalock et al., 2020; Diao et al., 2023). In LLMs, SparseLLM (Bai et al., 2024) approximates Eq. 2 through a sequence of block-level reconstruction constraints with iterative hidden-state updates to propagate pruning effects across blocks. LLM-Pruner (Ma et al., 2023) applies Eq. 3 for element-wise importance and explores structure-level aggregation, while LoRAPrune (Zhang et al., 2024) adapts it to LoRA for memory-efficient fine-tuning. Although higher-order derivatives can be included, prior work in both CNNs and LLMs (Ma et al., 2023; Molchanov et al., 2019; Zhang et al., 2024) shows that first-order information alone is sufficient for competitive results.

其中 $I_{W_i^j}$ 表示 θ 中元素的第 j 个估计重要性， H_{jj} 是 Hessian 矩阵的对角线。全局剪枝在较小的网络中已被广泛研究，例如卷积神经网络（CNN）、视觉变换器（Vision Transformers）和紧凑型语言模型（Molchanov 等，2016; Yang 等，2023; Diao 等，2023），其表现始终优于局部方法（Blalock 等，2020; Diao 等，2023）。在大语言模型（LLMs）中，SparseLLM（Bai 等，2024）通过一系列块级重建约束和迭代隐藏状态更新来近似公式 2，从而在块之间传播剪枝效果。LLM-Pruner（Ma 等，2023）应用公式 3 计算元素级重要性，并探索结构级聚合，而 LoRAPrune（Zhang 等，2024）则将其适应于 LoRA 以实现

内存高效的微调。尽管可以包含高阶导数，但 CNN 和 LLM 的先前研究（Ma 等，2023；Molchanov 等，2019；Zhang 等，2024）表明，仅使用一阶信息即可获得具有竞争力的结果。

Motivation. While local structured pruning is appealing for its efficiency, it remains fundamentally task-agnostic. These methods minimize layer-wise reconstruction errors to preserve input-output similarity with the dense model, which maintains perplexity and generic zero-shot behavior but does not optimize downstream accuracy. As shown in Table 1, we evaluate local methods on CMQA using Llama 2-13B under two calibration settings: a generic C4 corpus and task-specific CMQA samples. Even with task-informed calibration, the improvement is marginal, indicating that local pruning cannot effectively exploit task signals.

动机。尽管局部结构化剪枝因其高效性而具有吸引力，但它本质上仍然是任务无关的。这些方法通过最小化逐层重建误差来保持与密集模型的输入-输出相似性，这保持了困惑度和通用的零样本行为，但并未优化下游任务的准确性。如表 1 所示，我们在两种校准设置下使用 Llama 2-13B 在 CMQA 数据集上评估局部方法：通用的 C4 语料库和任务特定的 CMQA 样本。即使采用任务导向的校准，改进也十分有限，这表明局部剪枝无法有效利用任务信号。

In contrast, global pruning defines importance with respect to the overall model loss, naturally producing non-uniform sparsity patterns. Because its importance scores are computed on calibration data, global pruning can directly align pruning decisions with downstream objectives. This motivates our exploration of task-specific global iterative structured pruning, which unifies the efficiency of post-training methods with the flexibility to incorporate task-aware objectives.

相比之下，全局剪枝根据整个模型损失来定义重要性，自然地产生非均匀的稀疏模式。由于其重要性得分是在校准数据上计算的，全局剪枝可以直接将剪枝决策与下游目标对齐。这促使我们探索任务特定的全局迭代结构化剪枝，它将后训练方法的效率与能够融入任务感知目标的灵活性统一起来。

3 Method

3 方法

3.1 A Naive Case Study: One-shot Global Pruning

3.1 一个简单的案例研究：一次性全局剪枝

To assess the effectiveness and limitations of global pruning, we first replicate prior pruning protocols designed for smaller models (Frankle and Carbin,

为了评估全局剪枝的有效性及其局限性，我们首先复制了之前为较小模型设计的剪枝协议（Frankle 和 Carbin,

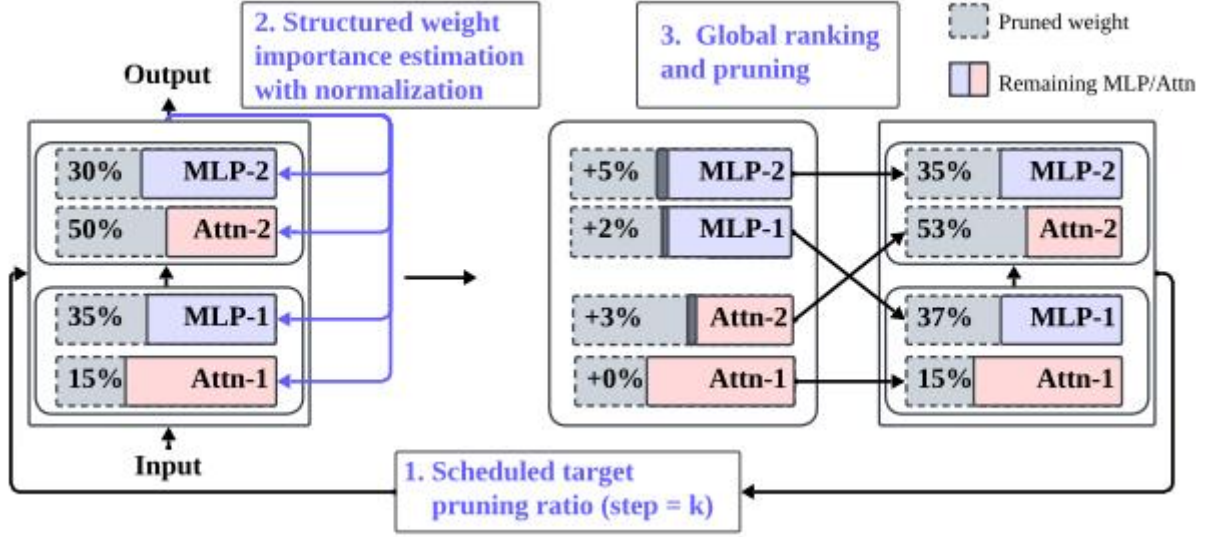


Figure 2: images/91112de1856487683d80ef63a855aaaaaf389cecd2cbe6c75e86cf8d8ea689c2.jpg

Figure 2: A detailed overview of the GISP. GISP performs iterative structured pruning guided by a ratio scheduler.

图 2: GISP 的详细概述。GISP 通过比率调度器进行迭代结构化剪枝。

2018; Mallya and Lazebnik, 2018). Specifically, we conduct the following procedure:

2018; Mallya 和 Lazebnik, 2018)。具体来说，我们执行以下步骤：

1. Given a calibration dataset $D = \{(x_i, y_i)\}_{i=1}^N$, we calculate the importance score of each weight element by using the first order term from eq. (3).
2. We construct the structure-wise importance scores by summing a group of importance scores of each weight element, where a group is defined as one head of the attention layer or a channel of the MLP layer. After the aggregation, a normalization process is conducted in consideration of group size among different structures, allowing all structural importance to be comparable across the model.
3. Given a current pruning ratio ρ , globally rank the normalized structure-wise scores and select the Top-K lowest-scoring structured weights for pruning.
4. 给定一个校准数据集 $D = \{(x_i, y_i)\}_{i=1}^N$ ，我们通过使用公式 (3) 中的一阶项来计算每个权重元素的重要性得分。
5. 我们通过将每个权重元素的重要性得分进行求和来构建结构级的重要性得分，其中一组被定义为注意力层的一个头或 MLP 层的一个通道。在聚合之后，考虑到不同结构之间的组大小，进行归一化处理，使得模型中所有结构的重要性可以相互比较。
6. 给定当前的剪枝比例 ρ ，对归一化后的结构级得分进行全局排序，并选择得分最低的 Top-K 结构权重进行剪枝。

Compared to prior pruning works on small models (Frankle and Carbin, 2018; Mallya and Lazeb-

nik, 2018), we make two changes: (1) these methods typically adopt a prune-then-fine-tune paradigm. In contrast, we evaluate performance in the setting of post-training pruning without fine-tuning, consistent with common local pruning practices for LLMs, as the computational and memory costs of fine-tuning after each pruning iteration are prohibitive, and (2) we observe that attention blocks exhibit substantially higher importance scores than MLP blocks (see Fig 3(c)), often by an order of magnitude. To address this imbalance, we normalize importance scores within attention and MLP blocks separately, which empirically yields improved accuracy.

与之前在小型模型上的剪枝工作 (Frankle 和 Carbin, 2018; Mallya 和 Lazebnik, 2018) 相比, 我们进行了两个改变: (1) 这些方法通常采用“剪枝-然后-微调”的范式。相反, 我们在不进行微调的情况下评估后训练剪枝的性能, 这与大型语言模型 (LLMs) 常见的本地剪枝实践一致, 因为每次剪枝迭代后进行微调的计算和内存成本是不可行的; (2) 我们观察到注意力块的重要性得分显著高于 MLP 块 (参见图 3(c)), 通常高出一个数量级。为了解决这种不平衡, 我们分别在注意力块和 MLP 块内部对重要性得分进行归一化处理, 这在实践中提高了准确性。

Empirical study. We perform experiments on one-shot global pruning and compare it with one representative structured local pruning method, Wanda. These experiments are conducted on Llama2-7B with a target pruning ratio of 20-50%. As shown in

实证研究。我们在单次全局剪枝上进行实验, 并将其与一种具有代表性的结构化局部剪枝方法 Wanda 进行比较。这些实验是在 Llama2-7B 模型上进行的, 目标剪枝比率为 20-50%。如所示

Table 1: Overall average CMQA accuracy (%) under different calibration datasets and pruning ratios. Local pruning methods have limited performance improvement, even with a task-informed calibration dataset.

表 1: 在不同的校准数据集和剪枝比例下, 整体平均 CMQA 准确率 (%). 局部剪枝方法即使使用任务感知的校准数据集, 性能提升也十分有限。

Method	Calibration Data	Pruning Ratio	AVG ACC
Wanda-sp	C4	20%	66.36
40%	58.24		
CMQA	20%	66.30	
40%	59.11		
LLM-Pruner	C4	20%	65.63
40%	55.06		
CMQA	20%	57.30	
40%	41.99		
FLAP	C4	20%	66.15
40%	61.73		
CMQA	20%	65.00	
40%	59.89		
OWL	C4	20%	66.62
40%	59.87		
CMQA	20%	66.98	
40%	60.84		

方法	校准数据	修剪比例	平均准确率
万达-sp	C4	20%	66.36

40%	58.24		
CMQA	20%	66.30	
40%	59.11		
LLM-Pruner	C4	20%	65.63
40%	55.06		
CMQA	20%	57.30	
40%	41.99		
FLAP	C4	20%	66.15
40%	61.73		
CMQA	20%	65.00	
40%	59.89		
OWL	C4	20%	66.62
40%	59.87		
CMQA	20%	66.98	
40%	60.84		

Table 2: Evaluation of one-shot global pruning (marked as one-shot GP) on perplexity (PPL) with C4 as the calibration dataset.

表 2: 使用 C4 作为校准数据集, 对一次性全局剪枝 (标记为 one-shot GP) 在困惑度 (PPL) 上的评估。

Method	Pruning ratio	Wiki2	PTB
Wanda-sp	20%	22.71	101.23
30%	35.43	138.41	
40%	51.85	185.09	
50%	81.47	218.31	
One-shot GP	20%	17.93	63.09
30%	26.99	81.48	
40%	53.45	151.80	
50%	159.47	353.55	

方法	修剪比例	Wiki2	PTB
万达-sp	20%	22.71	101.23
30%	35.43	138.41	
40%	51.85	185.09	
50%	81.47	218.31	
单次 GP	20%	17.93	63.09
30%	26.99	81.48	
40%	53.45	151.80	
50%	159.47	353.55	

Table 2, one-shot global pruning surpasses Wanda at low ratios but degrades at high sparsity, indicating that naive one-shot pruning is viable yet unstable, especially in high-pruning-ratio regions, which motivates our iterative strategy introduced next.

表 2 显示，一次性全局剪枝在较低剪枝比例下优于 Wanda，但在高稀疏度下性能下降，这表明简单的单次剪枝方法虽然可行但不够稳定，尤其是在高剪枝比例区域，这促使我们接下来引入迭代策略。

3.2 GISP: Global Iterative Structured Pruning

3.2 GISP: 全球迭代结构化剪枝

3.2.1 Stabilizing High-Ratio Pruning

3.2.1 高比例剪枝的稳定化

Motivation. We hypothesize that one potential issue of one-shot global pruning is that it removes a large portion of weights at once, increasing the risk of over-pruning important weights. A potential solution to this issue is iterative global pruning, which gradually prunes the model by applying a small pruning ratio in each step. This approach enables more precise identification of truly redundant weights, leveraging iterative feedback to refine pruning decisions.

动机。我们假设一次性全局剪枝的一个潜在问题在于，它一次性移除大量权重，增加了重要权重被过度剪枝的风险。解决这一问题的一个潜在方法是迭代式全局剪枝，即在每一步应用较小的剪枝比例，逐步对模型进行剪枝。这种方法能够更精确地识别真正冗余的权重，利用迭代反馈来优化剪枝决策。

Global Iterative Structured Pruning. Building upon the procedure detailed in Section 3.1, given a predefined number of iteration steps n and a target

全局迭代结构化剪枝。在第 3.1 节详细描述的流程基础上，给定预定义的迭代步数 n 和一个目标

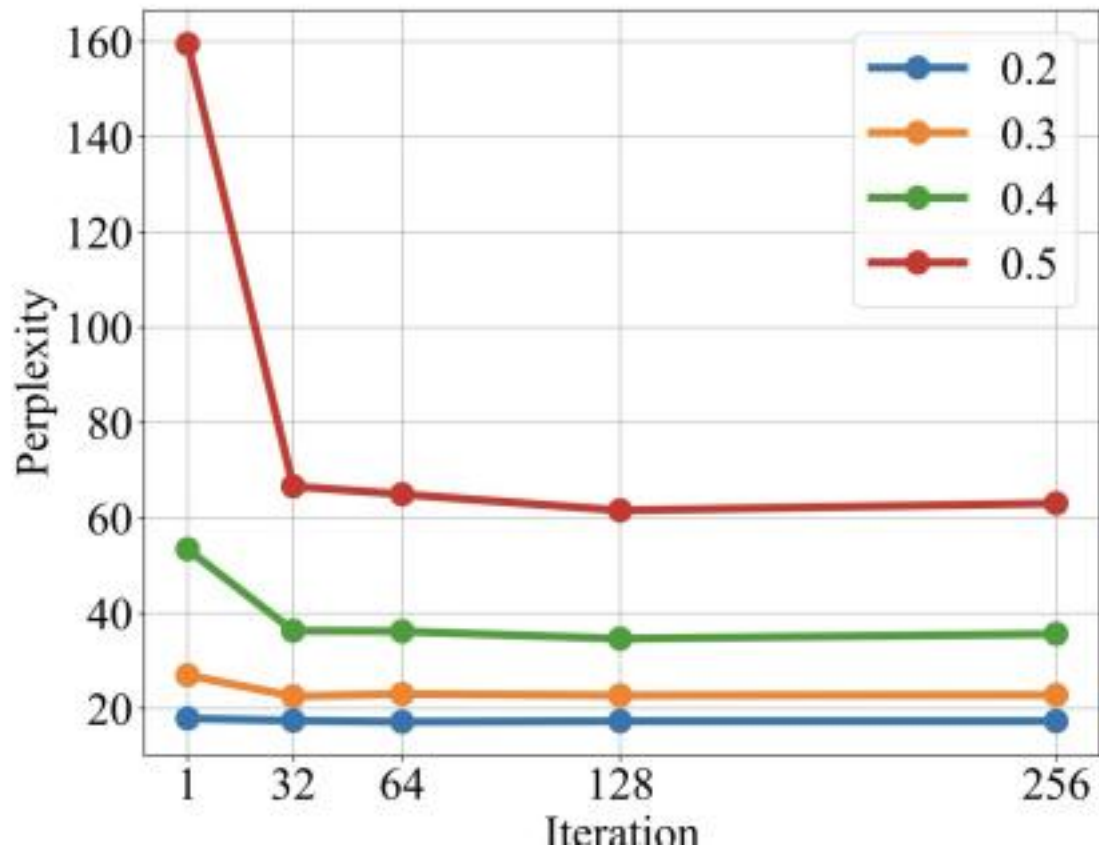


Figure 3: images/941b992eb243abef6c91dcbb837658c602e8f22aa526c81cefb17098a7ce8439.jpg

(a)

(a)

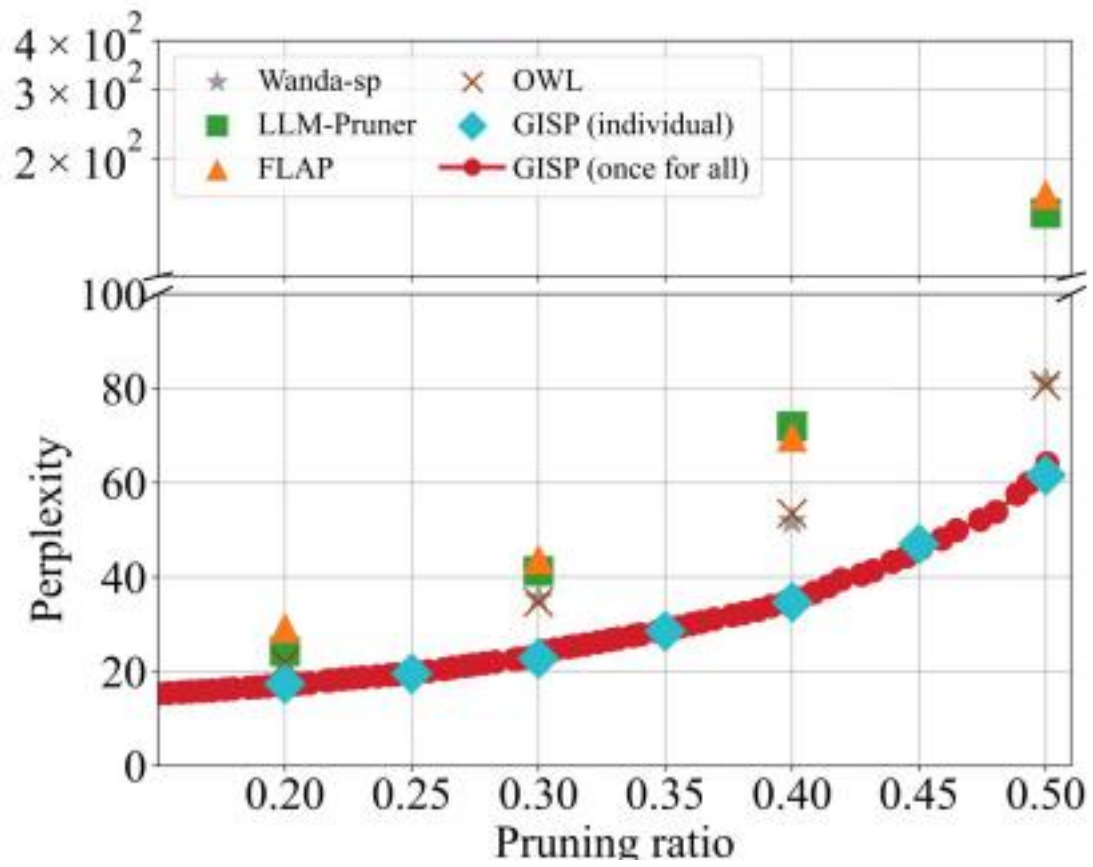


Figure 4: images/ba273848a64976706a8d948ebc714b9240c95c9813d104d18ed4f34e0c9e605e.jpg

(b)

(b)

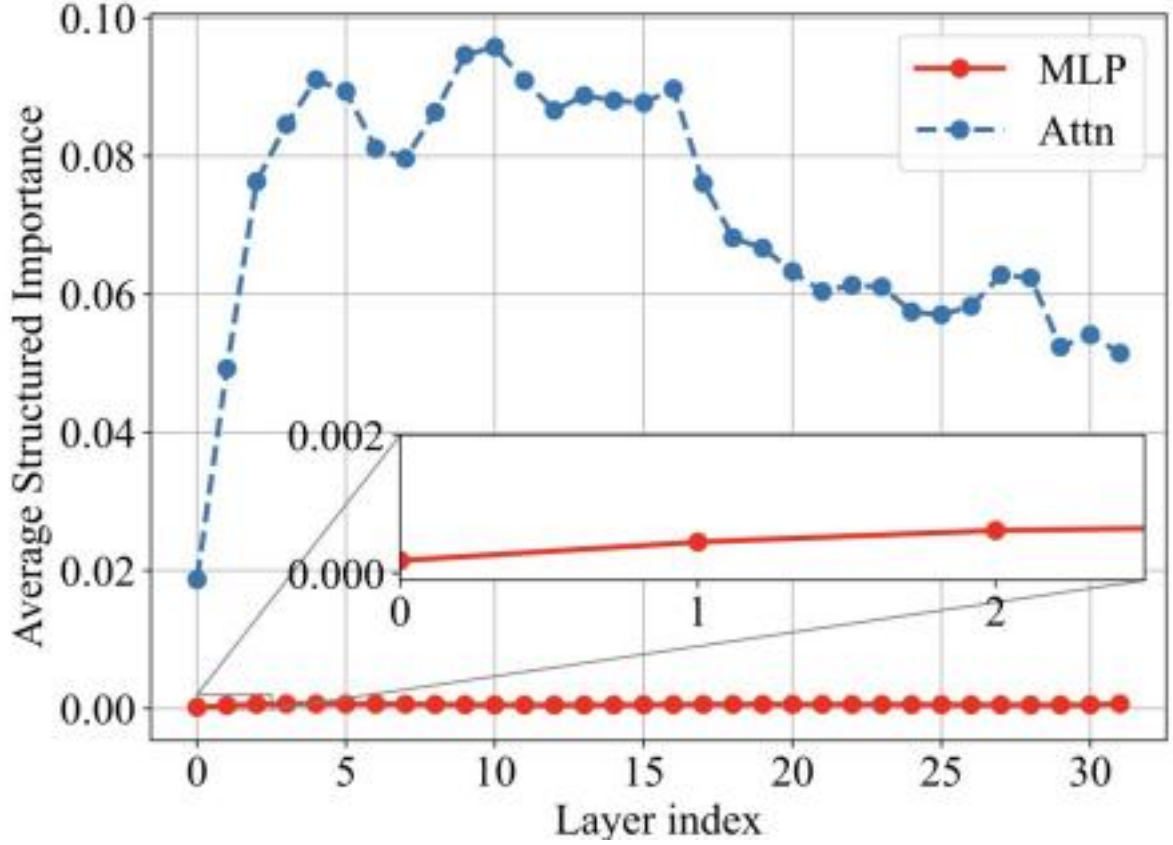


Figure 5: images/05e74a6e89b41e89e1ab58c2afd564da04c9f3a6856f3c6489264457c3f96a1c.jpg

(c)

Figure 3: (a) Perplexity analysis for various iteration settings. Iteration alleviates high-pruning-ratio perplexity collapse. (b) Perplexity analysis between GISP and other baselines. (c) Magnitude comparison between different types of structured weight importance.

(c)

图 3: (a) 不同迭代设置下的困惑度分析。迭代缓解了高剪枝率导致的困惑度崩溃。(b) GISP 与其他基线模型之间的困惑度分析。(c) 不同类型的结构化权重重要性之间的幅度比较。

Algorithm 1 Global Iterative Structured Pruning

算法 1 全局迭代结构化剪枝

Require: Network parameters θ_0 ; global target pruning ratio ρ ; iteration steps n ; calibration dataset $D = \{(x_i, y_i)\}_{i=1}^N$. Ensure: Pruned model parameters θ_n . $\{\rho_t\}_{t=1}^n \leftarrow \text{RATIOSCHEDULER}(\rho, n)$ for $k \leftarrow 1$ to n do $I(\theta_{k-1}) \leftarrow |\frac{\partial \mathcal{D}}{\partial \theta_{k-1}} \odot \theta_{k-1}| \triangleright$ First-order importance $I(\theta_{k-1}) \leftarrow \text{SUMAGGREGATE}(I(\theta_{k-1}))$ $I(\theta_{k-1}) \leftarrow \frac{I(\theta_{k-1})}{|\theta_{k-1}|} \triangleright$ Normalized Aggregation over heads/MLP channels $\tau_k \leftarrow \text{TOPK}(I(\theta_{k-1}), \rho_k)$ $\square$ Global pruning threshold across different structures at ratio ρ_k $m \leftarrow \mathbf{1}[I(\theta_{k-1}) > \tau_k]$ $\square$ Binary mask $\theta_k \leftarrow \theta_{k-1} \odot m$ $\square$ Apply pruning mask end for return θ_n

要求: 网络参数 θ_0 ; 全局目标剪枝比例 ρ ; 迭代步数 n ; 校准数据集 $D = \{(x_i, y_i)\}_{i=1}^N$ 。保证: 剪枝后的模型参数 θ_n 。 $\{\rho_t\}_{t=1}^n \leftarrow \text{RATIOSCHEDULER}(\rho, n)$ 对于 $k \leftarrow 1$ 到 n 循环执行 $I(\theta_{k-1}) \leftarrow |\frac{\partial \mathcal{D}}{\partial \theta_{k-1}} \odot \theta_{k-1}| \triangleright$ 一阶重要性 $I(\theta_{k-1}) \leftarrow \text{SUMAGGREGATE}(I(\theta_{k-1}))$ $I(\theta_{k-1}) \leftarrow \frac{I(\theta_{k-1})}{|\theta_{k-1}|} \triangleright$

头/MLP 通道上的归一化聚合 $\tau_k \leftarrow \text{TOPK}(I(\theta_{k-1}), \rho_k)$ $\square$ 不同结构上的全局剪枝阈值比例为 $\rho_k m \leftarrow \mathbf{1}[I(\theta_{k-1}) > \tau_k]$ $\square$ 二值掩码 $\theta_k \leftarrow \theta_{k-1} \odot m$ $\square$ 应用剪枝掩码 结束循环返回 θ_n

pruning ratio ρ , GISP performs pruning iteratively using a small pruning ratio ρ_k at iteration k , as shown in Fig 2. The detailed algorithm block is provided in Algorithm 1. To control pruning at each step, we use a linear scheduler that gradually increases the pruning ratio across iterations, ensuring that each iteration prunes the same number of structures. More details can be found in Sec. A.2

剪枝比例 ρ , GISP 在每次迭代中使用较小的剪枝比例 ρ_k 进行剪枝, 如图 2 所示。详细的算法流程图见算法 1。为了在每一步控制剪枝, 我们使用一个线性调度器, 逐步增加剪枝比例, 确保每次迭代剪枝的结构数量相同。更多细节请参见第 A.2 节。

Empirical study. For the iteration study, we vary the number of pruning steps (1, 32, 64, 128, and 256) across four target pruning ratios (20%, 30%, 40%, and 50%). For comparison with local pruning, we measure perplexity (PPL) and include four representative post-training structured pruning baselines: two uniform local pruning methods (Wanda and LLM-Pruner) and two non-uniform local pruning methods (FLAP and OWL). All experiments are performed on the Llama2-7B model using the C4 calibration dataset. The results are shown in Fig 3. We summarize our main findings below:

实证研究。对于迭代研究, 我们在四个目标剪枝比例 (20%、30%、40% 和 50%) 下变化剪枝步骤的数量 (1、32、64、128 和 256)。为了与本地剪枝进行比较, 我们测量了困惑度 (PPL), 并包括了四个具有代表性的后训练结构化剪枝基线: 两种均匀本地剪枝方法 (Wanda 和 LLM-Pruner) 以及两种非均匀本地剪枝方法 (FLAP 和 OWL)。所有实验均在 Llama2-7B 模型上使用 C4 校准数据集进行。结果如图 3 所示。我们总结主要发现如下:

1) Iteration is key to global pruning in the high-sparsity regime. From Fig 3(a) and (b), we first observe that introducing iterative pruning alleviates the issue of global pruning at a high pruning ratio:

1) 迭代是高稀疏度下全局剪枝的关键。从图 3(a) 和 (b) 中, 我们首先观察到引入迭代剪枝可以缓解在高剪枝比例下的全局剪枝问题:

Table 3: Pruning time comparison across methods. “Amortized time” divides the total time by the number of usable subnetworks produced.

表 3: 不同方法之间的剪枝时间比较。“平均时间”将总时间除以生成的可用子网络数量。

Method	Usable sparsities	Total time (min)	Amortized time (min)
Wanda-sp	4	7.10	1.78
OWL	4	13.90	3.48
LLM-Pruner	4	6.80	1.70
GISP (ours)	112	125.84	1.12

方法	可使用的稀疏性	总时间 (分钟)	摊销时间 (分钟)
万达-sp	4	7.10	1.78
OWL	4	13.90	3.48
LLM-Pruner	4	6.80	1.70
GISP (我们的)	112	125.84	1.12

even a coarse setting of 32 steps (equal to the layer count of Llama2-7B) is enough to cut the 50%-pruning-ratio PPL by 92.82. As a result, the GISP can consistently achieve lower PPL compared

to local pruning methods.

即使是一个粗略的 32 步设置（等于 Llama2-7B 的层数）也足以将 50% 剪枝率的 PPL 降低 92.82。因此，GISP 能够持续地比局部剪枝方法获得更低的 PPL。

2) Global iterative pruning outperforms local baselines at scale. With iteration, global pruning consistently achieves lower perplexity than local pruning methods across all sparsity regimes (Fig. 3(b)). This establishes that iteration not only stabilizes global pruning but also makes it competitive against strong local baselines in the post-training LLM setting. Crucially, these gains are obtained without intermediate recovery or fine-tuning, demonstrating that iteration alone is effective.

2) 全局迭代剪枝在大规模情况下优于局部基线方法。随着迭代的进行，全局剪枝在所有稀疏度范围内始终比局部剪枝方法获得更低的困惑度（图 3(b)）。这表明，迭代不仅能够稳定全局剪枝，还能使其在微调后的大型语言模型（LLM）设置中与强大的局部基线方法竞争。关键的是，这些收益是在不进行中间恢复或微调的情况下获得的，这表明仅靠迭代本身就非常有效。

3.2.2 Achieving “Once-for-All” and Amortizing the Iteration Cost

3.2.2 实现 “一次到位” 并摊销迭代成本

Iterative global pruning is computationally more demanding than local or one-shot pruning. Running such a computationally intensive procedure to obtain only a single subnetwork at a fixed sparsity level would be impractical in deployment. Table 3 compares the wall-clock pruning time of several structured pruning methods under our experimental setup. While GISP requires a substantially longer total runtime due to its iterative steps, the amortized cost per deployable subnetwork is comparable to, or even lower than, that of local methods once the once-for-all property is considered.

迭代式全局剪枝在计算上比局部剪枝或一次性剪枝更加耗费资源。在部署中，运行这种计算密集型的流程以仅获得一个固定稀疏度下的子网络是不切实际的。表 3 比较了在我们的实验设置下，几种结构化剪枝方法的墙上时间（实际运行时间）。虽然由于迭代步骤，GISP 需要显著更长的总运行时间，但一旦考虑 “一次生成所有” 特性，每个可部署子网络的平均成本与局部方法相当，甚至更低。

Moreover, in iterative global pruning, each step removes new self-attention heads and MLP channels based on the already-pruned model from the

此外，在迭代式全局剪枝中，每一步都基于已剪枝的模型，移除新的自注意力头和 MLP 通道。

previous step, naturally forming a nested sub-network structure (Cai et al., 2019). This nested property and computational cost from iteration motivated us to wonder:

上一步骤，自然形成一个嵌套的子网络结构（Cai 等，2019）。这种嵌套特性以及每次迭代的计算成本促使我们思考：

Can GISP enable once-for-all pruning? In other words, if we iteratively prune toward a high target ratio (e.g., 50%), can the intermediate subnetworks with lower pruning ratios (e.g., 20%, 30%) already perform well, thereby eliminating the need to conduct separate pruning runs for each individual pruning ratio?

GISP 是否能够实现一次性剪枝？换句话说，如果我们迭代地剪枝以达到较高的目标比例（例如 50%），中间子网络在较低的剪枝比例（例如 20%、30%）下是否已经能够表现良好，从而无需为每个单独的剪枝比例进行独立的剪枝运行？

To investigate this, we conduct a single iterative pruning procedure on Llama2-7B, targeting 50% sparsity over 112 iterations. We saved the intermediate pruned model at every step and evalu-

ated its PPL. The results are presented in Fig 3(b). The relationship between perplexity and the pruning ratio is remarkably smooth, indicating a stable and well-behaved pruning trajectory. Crucially, the performance of the intermediate models at different pruning ratios is on par with the performance of models generated from individual, shorter pruning runs (marked as “individual” variant) tuned specifically for those respective targets. To this end, this demonstrates the once-for-all capacity of GISP.

为了研究这一点，我们在 Llama2-7B 上进行了一次迭代剪枝过程，目标是在 112 次迭代中实现 50% 的稀疏度。我们在每一步保存了中间剪枝模型，并评估了其 PPL。结果如图 3(b) 所示。困惑度与剪枝比例之间的关系非常平滑，表明剪枝轨迹稳定且行为良好。重要的是，不同剪枝比例下的中间模型的性能与从单独、较短的剪枝运行生成的模型（标记为 “individual” 变体）的性能相当，这些模型是专门针对相应目标进行调优的。为此，这证明了 GISP 的一次剪枝即全部（once-for-all）能力。

It is important to note that this “once-for-all” capability is a unique advantage of the iterative global pruning. It enables practitioners to obtain an entire Pareto frontier of accuracy-vs-sparsity models from a single computational investment, offering immense practical flexibility. This property is not achievable with local pruning methods. As formulated in Eq. 1, local pruning is a layer-wise optimization that requires the target pruning ratio for each layer to be specified in advance. Consequently, creating models for different sparsity levels necessitates entirely separate and independent pruning runs.

需要注意的是，这种 “一次完成” 的能力是迭代式全局剪枝的独特优势。它使实践者能够通过一次计算投入，获得整个精度-稀疏性模型的帕累托前沿，从而提供巨大的实际灵活性。这种特性是局部剪枝方法所无法实现的。正如公式 1 中所表述的，局部剪枝是一种逐层优化方法，要求预先指定每一层的目标剪枝比例。因此，创建不同稀疏度级别的模型需要完全独立且分开的剪枝运行。

3.3 GISP as a Task-Specific Pruner

3.3 GISP 作为任务特定的剪枝器

As discussed in Sec. 2, local pruning remains task-agnostic because its layer-wise reconstruction objective cannot align with downstream goals, even when calibration data carries task-specific information. In contrast, global pruning defines importance with respect to a model-level loss, offering the potential for task alignment. We now instantiate and validate this property in GISP.

如第 2 节所述，局部剪枝仍然是任务无关的，因为其逐层重建目标无法与下游目标对齐，即使校准数据包含任务特定的信息。相反，全局剪枝是基于模型级损失来定义重要性的，从而具有任务对齐的潜力。我们现在在 GISP 中实例化并验证这一特性。

Objective-level formulation. Because GISP evaluates importance with respect to a model-level loss (Eq. 3), we can instantiate a task-aligned objective by replacing the loss in Eq. 2 with a task-specific target L_{task} . Our importance reduces to the same first-order form with a different objective:

目标级的表述。由于 GISP 是根据模型级损失（公式 3）来评估重要性，我们可以通过将公式 2 中的损失替换为特定任务的目标 L_{task} 来实例化一个与任务对齐的目标。我们的重要性减少为具有不同目标的同一阶形式：

$$I_W = |\langle \nabla_W L_{task}, W \rangle|. \quad (4)$$

This simple substitution turns GISP into a task-specific pruner while remaining post-training and structure-aware.

这种简单的替换将 GISP 转换为一个特定任务的剪枝器，同时保持其在训练后和结构感知的特性。

Two instantiations. We consider two common families of L_{task} that match our evaluation tasks:

两个实例。我们考虑两个常见的 L_{task} 家族，它们与我们的评估任务相匹配：

(i) Perplexity loss for text generation (language modeling), where $L_{task} = token-levelcross-entropy$ on an open-domain (e.g., C4) or in-domain (e.g., GSM8K) corpus; To be specific, the importance metrics are obtained from objective:

(i) 文本生成（语言建模）的困惑度损失，其中 $L_{task} = token - levelcross - entropy$ 在开放域（例如 C4）或领域内（例如 GSM8K）语料库上；具体而言，重要性指标来源于以下目标：

$$L = -\frac{1}{N} \sum_{i=1}^N \log p(x_i | x_{<i}) \quad (5)$$

where L is the loss function, N is the number of tokens and $p(x_i | x_{<i})$ is the probability of token x_i given all previous tokens.

其中 L 是损失函数， N 是 token 的数量， $p(x_i | x_{<i})$ 是在所有先前 token 的条件下，token x_i 的概率。

(ii) Margin loss for decision-oriented, multi-option QA. For example, the CMQA dataset differs from pure language modeling in that each question is paired with one correct (positive) and multiple incorrect (negative) answers. During inference, the model ranks each ‘Question + Answer’ pair by perplexity and selects the answer with the lowest score. Simply minimizing perplexity on positive answers is insufficient, as pruning may disproportionately reduce the loss of negative candidates relative to the correct one, causing the model to choose an incorrect answer even if the correct answer’s loss remains largely unchanged. In other words, the actual factor of classification performance is the model’s ability to distinguish correct from incorrect answers (the decision boundary).

(ii) 针对面向决策的多选项问答的边距损失。例如，CMQA 数据集与纯粹的语言建模不同，每个问题都与一个正确的（正类）和多个错误的（负类）答案配对。在推理过程中，模型通过困惑度对每个“问题 + 答案”对进行排序，并选择得分最低的答案。仅仅最小化正类答案的困惑度是不够的，因为剪枝可能会相对于正确答案不成比例地减少负类候选答案的损失，导致模型即使正确答案的损失基本不变，仍可能选择错误的答案。换句话说，实际影响分类性能的因素是模型区分正确答案和错误答案的能力（决策边界）。

To preserve the model’s decision boundary, we define a margin-based importance using a task-formatted calibration set:

为了保持模型的决策边界，我们使用任务格式校准集定义基于边距的重要性：

$$I_{W_i^j} = \left| \left(\frac{\partial L_+}{\partial W_i^j} - \frac{\partial L_-}{\partial W_i^j} \right) W_i^j \right| \quad (6)$$

Where L_+ denotes the average loss on positive labels and L_- denotes the average loss on negative labels. Intuitively, Eq. (6) preserves the loss gap between correct and incorrect candidates, aligning pruning with task decisions. We will examine the effectiveness of GISP as a task-specific pruner in Sec. 4.2. Importantly, such a transition from a perplexity-based loss to a task-specific loss is not feasible for local pruning methods, which rely on layer-wise MSE loss for importance estimation.

其中， L_+ 表示正样本上的平均损失， L_- 表示负样本上的平均损失。直观来看，式（6）保持了正确候选与错误候选之间的损失差距，使剪枝与任务决策保持一致。我们在第 4.2 节将探讨 GISP 作为任务特定剪枝器的有效性。重要的是，从基于困惑度的损失向任务特定损失的转变，对于依赖逐层均方误差损失进行重要性估计的局部剪枝方法来说是不可行的。

Table 4: Comparison of different pruning methods on perplexity and CMQA downstream reason-

ing accuracy.

表 4：不同剪枝方法在困惑度和 CMQA 下游推理准确性上的比较。

Pruning Ratio	Method	Perplexity on Wikitext2 $\downarrow$		Downstream ACC (%) $\uparrow$	
Llama2		Llama2			
7B	13B	7B	13B		
0%	Dense	12.19	10.98	66.68	69.19
20%	Wanda-sp	22.71	14.64	61.77	66.30
LLM-Pruner	24.25	19.99	50.95	57.30	
FLAP	29.19	16.95	61.27	65.00	
ShortGPT	43.88	19.95	55.75	60.84	
OWL	21.80	14.76	62.64	66.98	
GISP (ours)	17.01	15.10	63.46	67.61	
30%	Wanda-sp	35.43	19.73	57.14	62.69
LLM-Pruner	41.24	28.47	41.37	46.26	
FLAP	43.75	21.32	56.90	63.28	
ShortGPT	126.42	84.84	50.01	56.86	
OWL	34.64	19.02	58.33	63.27	
GISP (ours)	24.27	19.53	60.68	66.12	
40%	Wanda-sp	51.85	32.91	50.12	59.11
LLM-Pruner	71.93	50.01	39.13	41.99	
FLAP	69.64	37.76	53.01	59.89	
ShortGPT	189.17	92.38	45.35	48.73	
OWL	53.47	31.13	51.50	60.84	
GISP (ours)	34.54	26.56	55.28	63.34	
50%	Wanda-sp	81.47	64.17	43.52	51.60
LLM-Pruner	144.99	86.34	38.62	40.92	
FLAP	161.84	66.38	47.84	56.29	
ShortGPT	387.94	276.08	41.75	41.75	
OWL	80.59	65.28	44.82	54.17	
GISP (ours)	64.07	42.07	48.54	57.50	

剪枝比例	方法	维基文本 2 的困惑度 $\downarrow$		下游 ACC (%) $\uparrow$	
Llama2		Llama2			
7B	13B	7B	13B		
0%	密集	12.19	10.98	66.68	69.19
20%	Wanda-sp	22.71	14.64	61.77	66.30
LLM-Pruner	24.25	19.99	50.95	57.30	
FLAP	29.19	16.95	61.27	65.00	
短 GPT	43.88	19.95	55.75	60.84	
OWL	21.80	14.76	62.64	66.98	

GISP (我们的)	17.01	15.10	63.46	67.61	
30%	万达-斯帕	35.43	19.73	57.14	62.69
LLM-Pruner	41.24	28.47	41.37	46.26	
FLAP	43.75	21.32	56.90	63.28	
短 GPT	126.42	84.84	50.01	56.86	
OWL	34.64	19.02	58.33	63.27	
GISP (我们的)	24.27	19.53	60.68	66.12	
40%	万达-斯帕	51.85	32.91	50.12	59.11
LLM-Pruner	71.93	50.01	39.13	41.99	
FLAP	69.64	37.76	53.01	59.89	
短 GPT	189.17	92.38	45.35	48.73	
OWL	53.47	31.13	51.50	60.84	
GISP (我们的)	34.54	26.56	55.28	63.34	
50%	万达-sp	81.47	64.17	43.52	51.60
LLM-Pruner	144.99	86.34	38.62	40.92	
FLAP	161.84	66.38	47.84	56.29	
短 GPT	387.94	276.08	41.75	41.75	
OWL	80.59	65.28	44.82	54.17	
GISP (我们的)	64.07	42.07	48.54	57.50	

4 Experiments

4 实验

Models and Evaluation. We evaluate GISP on the popular Llama2-7B/13B (Touvron et al., 2023b), Llama3-8B (Grattafiori et al., 2024), Mistral-0.3-7B (Jiang et al., 2023), and two reasoning model DeepSeek-R1-Distill-Llama-3-8B (DeepSeek-AI et al., 2025) and Qwen3-8B (Yang et al., 2025a). Following previous work (Ma et al., 2023; An et al., 2023), we evaluate the pruned model on three categories of tasks: the perplexity metric on Wikitext2 (Merity et al., 2016) text generation, exact-match accuracy on math benchmark GSM8K (Cobbe et al., 2021) that require step-by-step reasoning, and post-training accuracy on commonsense reasoning (CMQA), which includes BoolQ (Clark et al., 2019), PIQA (Bisk et al., 2020), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2019), ARC-Easy (Clark et al., 2018), ARC-Challenge (Clark et al., 2018), and OpenbookQA (Mihaylov et al., 2018). We report average CMQA accuracy in this section, and detailed task-wise accuracy in Sec. A.7. Additionally, we further evaluate the proposed GISP on the domain-specific MedQA benchmark (Jin et al., 2020), including medical exam-style multiple-choice questions. The result can be found in Sec. A.5. For calibration, we use different datasets for perplexity evaluation and downstream accuracy: 1) for perplexity evaluation on WikiText-2, we follow the standard setup and use C4 for calibration, 2) for downstream accu

模型与评估。我们在流行的 Llama2-7B/13B (Touvron 等, 2023b)、Llama3-8B (Grattafiori 等, 2024)、Mistral-0.3-7B (Jiang 等, 2023) 以及两个推理模型 DeepSeek-R1-Distill-Llama-3-8B (DeepSeek-AI 等, 2025) 和 Qwen3-8B (Yang 等, 2025a) 上评估 GISP。按照之前的工作 (Ma 等, 2023; An 等, 2023), 我们在三个任务类别上评估剪枝后的模型: 在 Wikitext2 (Merity 等, 2016) 文本生成任务上的困惑度指标、在需要逐步推理的数学基准 GSM8K (Cobbe 等, 2021) 上的精确匹配准确率, 以及在常识推理 (CMQA) 上的后训练准确率, 其中包括 BoolQ (Clark 等, 2019)、PIQA (Bisk 等, 2020)、HellaSwag (Zellers 等, 2019)、WinoGrande (Sakaguchi 等, 2019)、ARC-Easy (Clark 等, 2018)、ARC-Challenge (Clark 等, 2018) 和 OpenbookQA (Mihaylov 等, 2018)。我们在本节中报告平均的

CMQA 准确率，并在 Sec. A.7 中详细列出每个任务的准确率。此外，我们还进一步在领域特定的 MedQA 基准（Jin 等，2020）上评估所提出的 GISP，包括医学考试风格的多项选择题。结果可在 Sec. A.5 中找到。对于校准，我们使用不同的数据集进行困惑度评估和下游准确率评估：1) 在 WikiText-2 上进行困惑度评估时，我们遵循标准设置并使用 C4 进行校准，2) 在下游准确率评估时，我们使用不同的数据集。

Table 5: Comparison of different pruning methods for advanced models.

表 5：高级模型不同剪枝方法的比较。

Pruning Ratio	Method	Perplexity on Wikitext2 $\downarrow$		Downstream ACC (%) $\uparrow$	
Llama3 8B	Mistral-0.3 7B	Llama3 8B	Mistral-0.3 7B		
0%	Dense	14.14	15.14	69.99	70.47
20%	Wanda-sp	29.92	20.42	57.45	64.39
LLM-Pruner	23.21	\	56.51	\	
ShortGPT	118.62	52.74	57.68	57.75	
OWL	29.49	19.98	59.95	65.87	
GISP (ours)	24.18	18.17	65.28	66.60	
30%	Wanda-sp	48.83	32.61	52.03	58.24
LLM-Pruner	37.78	\	47.46	\	
ShortGPT	3972.28	599.82	43.53	41.10	
OWL	47.90	31.82	52.24	58.54	
GISP (ours)	31.73	25.58	59.66	63.48	
40%	Wanda-sp	81.67	55.41	43.61	51.89
LLM-Pruner	67.58	\	41.82	\	
ShortGPT	1576.47	909.21	43.37	39.68	
OWL	87.01	47.85	44.87	54.36	
GISP (ours)	46.10	34.31	53.51	58.30	
50%	Wanda-sp	133.29	79.41	41.32	44.38
LLM-Pruner	125.91	\	39.67	\	
ShortGPT	4135.73	1091.73	41.19	38.73	
OWL	130.77	76.20	41.86	46.09	
GISP (ours)	79.42	58.16	45.68	49.79	

修剪比例	方法	维基文本 2 的困惑度 $\downarrow$		下游 ACC (%) $\uparrow$	
Llama3 8B	Mistral-0.3 7B	Llama3 8B	Mistral-0.3 7B		
0%	密集	14.14	15.14	69.99	70.47
20%	万达-sp	29.92	20.42	57.45	64.39
LLM-Pruner	23.21	\	56.51	\	
短 GPT	118.62	52.74	57.68	57.75	
OWL	29.49	19.98	59.95	65.87	
GISP (我们的)	24.18	18.17	65.28	66.60	
30%	万达-sp	48.83	32.61	52.03	58.24
LLM-Pruner	37.78	\	47.46	\	

短 GPT	3972.28	599.82	43.53	41.10	
OWL	47.90	31.82	52.24	58.54	
GISP (我们的)	31.73	25.58	59.66	63.48	
40%	Wanda-sp	81.67	55.41	43.61	51.89
LLM-Pruner	67.58	\	41.82	\	
短 GPT	1576.47	909.21	43.37	39.68	
OWL	87.01	47.85	44.87	54.36	
GISP (我们的)	46.10	34.31	53.51	58.30	
50%	Wanda-sp	133.29	79.41	41.32	44.38
LLM-Pruner	125.91	\	39.67	\	
短 GPT	4135.73	1091.73	41.19	38.73	
OWL	130.77	76.20	41.86	46.09	
GISP (我们的)	79.42	58.16	45.68	49.79	

racy, we instead use task-specific calibration data. Specifically, in CMQA, we adopt a unified multi-task pruning setting by constructing a single calibration set from the training splits of all seven tasks under a shared 512K-token budget, evenly allocated across tasks. One shared model is pruned using this combined calibration set and then evaluated on each task. For each sampled question, we retain the task-formatted answer candidates. Under the margin-based objective, the correct answer is treated as positive, and the remaining candidates are treated as negatives. For GSM8K, we consider both C4 and the GSM8K training set for calibration. In particular, we sample 500 sequences from C4, each of length 2048 tokens, and 500 randomly sampled question-solution pairs from the GSM8K training set.

为了进行校准，我们改用任务特定的校准数据。具体而言，在 CMQA 中，我们采用统一的多任务剪枝设置，通过在共享的 512K-token 预算下，从七个任务的所有训练分割中构建一个单一的校准集，并将预算均匀分配给各个任务。然后，使用这个组合的校准集对一个共享模型进行剪枝，并在每个任务上进行评估。对于每个采样的问题，我们保留任务格式的答案候选。在基于边距的目标下，正确答案被视为正样本，其余候选则被视为负样本。对于 GSM8K，我们同时考虑 C4 和 GSM8K 训练集进行校准。具体来说，我们从 C4 中采样 500 个序列，每个序列长度为 2048 个 token，并从 GSM8K 训练集中随机采样 500 个问题-解答对。

Baselines. We compare GISP with four local pruning approaches in two main categories: (1) local uniform baselines, including Wanda-sp (Sun et al., 2024; An et al., 2023), LLM-Pruner (Ma et al., 2023); and (2) local non-uniform baselines: FLAP (An et al., 2023), OWL (Yin et al., 2025) on Wanda-sp. Additionally, we compare against a layer-wise pruning approach, ShortGPT (Men et al., 2024). We also include SparseLLM (Bai et al., 2024), a global pruning method originally proposed for unstructured pruning, by implementing a structured variant for comparison under our setting; the detailed results are provided in Appendix A.4. The iteration step in GISP is set to 112 in models with a 7-8B scale, and 280 in 13B to maintain the close iteration stride with these smaller variants. The detailed experimental setup

基线。我们在两个主要类别中将 GISP 与四种局部剪枝方法进行比较：(1) 局部均匀基线，包括 Wanda-sp (Sun 等，2024; An 等，2023)、LLM-Pruner (Ma 等，2023)；以及 (2) 局部非均匀基线：FLAP (An 等，2023)、OWL (Yin 等，2025) 在 Wanda-sp 上。此外，我们还与一种分层剪枝方法 ShortGPT (Men 等，2024) 进行比较。我们还包含 SparseLLM (Bai 等，2024)，这是一种最初为非结构化剪枝提出全局剪枝方法，通过实现一种结构化变体以在我们的设置下进行比较；详细结果见附录 A.4。在 7-8B 规模的模型中，GISP 的迭代步长设置为 112，在 13B 规模的模型中设置为 280，以保持与这些较小变体相近的迭代步长。详细的实验设置

Table 6: CMQA Accuracy of GISP on all seven tasks under different calibration datasets and prun-

ing ratios. The best results are marked in bold.

表 6: GISP 在不同校准数据集和剪枝比例下在所有七个任务上的 CMQA 准确率。最佳结果用粗体标出。

Calibration Dataset	Pruning Ratio	BoolQ	PIQA	Hellaswag	WinoGrande	ARC-E	ARC-C	OBQA	AVG
C4 + Perplexity	20%	73.30	78.45	73.09	68.11	67.59	42.92	42.00	63.64
	30%	69.20	76.71	69.68	65.98	37.46	40.80	60.36	
	40%	65.14	73.67	62.80	61.64	33.87	36.80	55.49	
	50%	58.41	68.28	50.79	58.64	28.41	32.20	48.68	
	CMQA + Perplexity	20%	80.80	77.86	76.39	71.82	74.75	46.33	67.16
CMQA + Margin	30%	80.83	75.52	71.91	69.69	44.71	41.60	65.21	
	40%	79.54	72.52	63.36	67.88	41.81	39.20	61.74	
	50%	76.18	67.52	51.67	61.56	36.77	36.40	55.65	
	CMQA + Margin	20%	80.28	79.00	76.83	72.22	75.17	43.80	67.61
	30%	81.16	76.77	72.87	71.59	45.73	41.80	66.12	
	40%	80.00	73.83	65.79	70.09	43.52	40.00	63.34	
	50%	72.97	69.91	55.15	65.59	38.23	37.60	57.50	

校准数据集	剪枝比例	布尔问答 (BoolQ)	PIQA	Hellaswag	WinoGrande	ARC-E	ARC-C	OBQA
C4 + Perplexity	20%	73.30	78.45	73.09	68.11	67.59	42.92	42.00
30%	69.20	76.71	69.68	65.98	62.67	37.46	40.80	60.36
40%	65.14	73.67	62.80	61.64	54.55	33.87	36.80	55.49
50%	58.41	68.28	50.79	58.64	44.02	28.41	32.20	48.68
CMQA + Perplexity	20%	80.80	77.86	76.39	71.82	74.75	46.33	42.20
30%	80.83	75.52	71.91	69.69	72.22	44.71	41.60	65.21
40%	79.54	72.52	63.36	67.88	67.85	41.81	39.20	61.74
50%	76.18	67.52	51.67	61.56	59.47	36.77	36.40	55.65
CMQA + Margin	20%	80.28	79.00	76.83	72.22	75.17	45.99	43.80
30%	81.16	76.77	72.87	71.59	72.94	45.73	41.80	66.12
40%	80.00	73.83	65.79	70.09	70.16	43.52	40.00	63.34
50%	72.97	69.91	55.15	65.59	63.09	38.23	37.60	57.50

is described in the Sec. A.3.

见第 A.3 节。

4.1 Experimental Results

4.1 实验结果

Perplexity of text generation tasks. Table 4 and Table 5 present the experimental results on the perplexity (PPL) of WikiText2 across four target pruning ratio levels. First of all, compared to the five baselines on dense Llama2-7B, 13B models, GISP achieves a clear lower PPL in most cases. Specifically, the improvement is particularly significant at the higher sparsity level (e.g., 40%, and 50%). Moreover, for multi-query attention-based LLMs such as Llama3-8B and Mistral-0.3, we observe the same consistent trend¹.

文本生成任务的困惑度。表 4 和表 5 展示了在四个目标剪枝比率水平下，WikiText2 的困惑度（PPL）的实验结果。首先，与在密集 Llama2-7B、13B 模型上的五个基线相比，GISP 在大多数情况下实现了更明显的较低 PPL。具体而言，在较高的稀疏度水平（例如 40% 和 50%）下，改进尤为显著。此外，对于基于多查询注意力的 LLM，如 Llama3-8B 和 Mistral-0.3，我们观察到了相同的一致趋势¹。

Downstream accuracy of commonsense reasoning tasks. Table 4 and Table 5 summarize the accuracy results of CMQA under downstream task evaluations. Note that Downstream Accuracy is evaluated by using CMQA calibration. We observe that on downstream tasks, GISP consistently achieves higher accuracy across all models and pruning ratios, with particularly strong gains at higher pruning levels, indicating its strength as a task-specific pruner.

常识推理任务的下游准确性。表 4 和表 5 总结了在下游任务评估下 CMQA 的准确性结果。请注意，下游准确性是通过使用 CMQA 校准进行评估的。我们观察到，在下游任务中，GISP 在所有模型和剪枝比率下都始终保持较高的准确性，尤其是在较高的剪枝级别下表现出特别显著的提升，这表明其作为特定任务剪枝器的优势。

Exact-match accuracy of answer-generation tasks. While CMQA evaluates multiple-choice reasoning, we further validate GISP on the arithmetic reasoning benchmark GSM8K, which follows a text-generation format and is evaluated by the presence of the gold answer in the generated output (Gold ACC). Table 7 compares different pruning methods and calibration datasets under 8-shot evaluation on DeepSeek-R1-Distill-Llama-3-8B and Qwen3-8B. The same trend holds across both models: task-informed calibration sub-

答案生成任务的精确匹配准确率。虽然 CMQA 评估多项选择推理，但我们进一步在算术推理基准 GSM8K 上验证 GISP，该基准采用文本生成格式，并通过生成输出中包含黄金答案（Gold ACC）来评估。表 7 比较了在 DeepSeek-R1-Distill-Llama-3-8B 和 Qwen3-8B 上进行 8-shot 评估的不同剪枝方法和校准数据集。两种模型均表现出相同的趋势：任务引导的校准子集

Table 7: Comparison of different pruning methods on GSM8K (8-shot) Gold ACC (%).

表 7：不同剪枝方法在 GSM8K（8-shot）上的黄金准确率（%）。

Model	Calibration	Method	Ratio	Gold ACC (%)
DeepSeek-R1-Distill-Llama-3-8B	N/A	Dense	0%	73.54%
C4	Wanda-sp	20%	0.00%	
GISP	20%	25.25%		
30%	14.33%			
40%	5.46%			
GSM8K	Wanda-sp	20%	29.19%	
GISP	20%	67.93%		
30%	50.80%			
40%	31.84%			

Qwen3-8B	N/A	Dense	0%	96.74%
C4	Wanda-sp	20%	25.40%	
30%	2.81%			
GISP	20%	47.31%		
30%	9.25%			
GSM8K	Wanda-sp	20%	85.90%	
30%	16.38%			
GISP	20%	90.52%		
30%	79.68%			

模型	校准	方法	比率	黄金 ACC
DeepSeek-R1-Distill-Llama-3-8B	N/A	密集	0%	73.54%
C4	万达-sp	20%	0.00%	
GISP	20%	25.25%		
30%	14.33%			
40%	5.46%			
GSM8K	万达-斯帕	20%	29.19%	
GISP	20%	67.93%		
30%	50.80%			
40%	31.84%			
通义千问 3-8B	N/A	密集	0%	96.74%
C4	万达-sp	20%	25.40%	
30%	2.81%			
地理信息系统专业人员 (Geographic Information Systems Professional)	20%	47.31%		
30%	9.25%			
GSM8K	万达-斯帕	20%	85.90%	
30%	16.38%			
GISP	20%	90.52%		
30%	79.68%			

stantially improves GISP over generic C4 calibration, and GISP consistently achieves stronger Gold ACC than Wanda-sp at the same sparsity levels. These results further support the effectiveness of task-aligned calibration for generative reasoning tasks. Notably, pruning reasoning-oriented LLMs remains challenging for current methods (Zhang et al., 2025; Sui et al., 2025); under default C4 calibration, Wanda-sp even collapses to 0% accuracy at 20% pruning on DeepSeek-R1-Distill-Llama-3-8B.

显著提升了通用 C4 校准的 GISP 性能，并且在相同稀疏度水平下，GISP 一直比 Wanda-sp 实现更强的黄金 ACC。这些结果进一步支持了任务对齐校准在生成式推理任务中的有效性。值得注意的是，当前方法在剪枝以推理为导向的 LLM 时仍然具有挑战性 (Zhang 等, 2025; Sui 等, 2025); 在默认 C4 校准下，Wanda-sp 在 DeepSeek-R1-Distill-Llama-3-8B 上甚至在 20% 剪枝时准确率会下降到 0%。

4.2 Task-specific Property of GISP

4.2 GISP 的任务特定属性

Ablation across seven CMQA tasks. We use CMQA to examine the task-specific property of GISP in a unified multi-task pruning setting described at the beginning of Sec. 4. We report detailed results on all tasks across pruning ratios of 20% to 50%. Table 6 summarizes two consistent trends: (1) Task-informed calibration helps even with a perplexity target: replacing C4 with CMQA data under the same perplexity objective yields gains at all ratios, indicating that GISP is an

在七个 CMQA 任务上的消融实验。我们使用 CMQA 在第 4 节开头描述的统一多任务剪枝设置中，考察 GISP 的任务特定属性。我们在剪枝比例为 20% 到 50% 的所有任务上报告详细结果。表 6 总结了两个一致的趋势：(1) 带任务信息的校准即使在困惑度目标下也有帮助：在相同困惑度目标下，用 CMQA 数据替换 C4 数据，在所有比例下都取得了收益，这表明 GISP 是一个

Table 8: Average CMQA accuracy after LoRA fine-tuning on an additional 2M CMQA samples.

表 8: 在额外 2M 个 CMQA 样本上进行 LoRA 微调后的平均 CMQA 准确率。

Method	20% sparsity		40% sparsity	
Before	After	Before	After	
Wanda-sp	61.84	67.43	50.09	58.66
GISP (ours)	63.46	68.93	55.28	60.52

方法	20% 稀疏性		40% 稀疏性	
之前	之后	之前	之后	
万达-斯帕	61.84	67.43	50.09	58.66
GISP (我们的)	63.46	68.93	55.28	60.52

Table 9: CMQA ACC Comparison between limited (1%) and default (100%) budgets on Llama2-7B.

表 9: Llama2-7B 上有限预算（1%）与默认预算（100%）的 CMQA 准确率对比。

Token budget	20%	30%	40%	50%
5,120 (1%)	63.86	60.46	55.83	49.20
512,000 (100%)	63.46	60.68	55.28	48.54
Absolute diff.	0.40	0.22	0.55	0.66

Token budget	20%	30%	40%	50%
5,120 (1%)	63.86	60.46	55.83	49.20
512,000 (100%)	63.46	60.68	55.28	48.54
绝对差值。	0.40	0.22	0.55	0.66

intrinsic task-specific pruner that can actively benefit from task signals from the calibration dataset. (2)

一种能够主动利用校准数据集中的任务信号来获益的内在任务特定剪枝器。(2)

Task-specific loss target brings further improvements: switching from perplexity to the proposed margin objective (Eq. 6) provides additional, consistent accuracy gains, especially at higher pruning ratios. These trends hold across tasks, showing that GISP can capture cross-task sensitivity within a shared structured subnetwork and supporting GISP as a practical task-specific pruner.

任务特定的损失目标带来进一步的改进：从困惑度切换到所提出的边距目标（公式 6）可以带来额外且一致的准确性提升，尤其是在较高的剪枝比率下。这些趋势在不同任务中均成立，表明 GISP 能够在共享的结构化子网络中捕捉跨任务的敏感性，并支持 GISP 作为一种实用的任务特定剪枝方法。

Post-pruning fine-tuning performance. We also examine whether the task-aligned subnetworks identified by GISP remain favorable for further downstream adaptation. Starting from the 20% and 40% pruned Llama2-7B checkpoints, we fine-tune the pruned models with LoRA on an additional 2M CMQA samples under identical configurations. Table 8 shows that GISP consistently achieves higher final accuracy after LoRA adaptation at both sparsity levels. This suggests that GISP not only yields stronger pruned subnetworks but also provides a better starting point for subsequent task adaptation.

剪枝后的微调性能。我们还检查通过 GISP 识别的任务对齐子网络是否在进一步下游适应中仍然具有优势。从 20% 和 40% 剪枝的 Llama2-7B 检查点开始，我们在相同的配置下，使用 LoRA 在额外的 2M CMQA 样本上对剪枝模型进行微调。表 8 显示，GISP 在两种稀疏级别下经过 LoRA 适应后始终获得更高的最终准确率。这表明 GISP 不仅能够生成更强的剪枝子网络，还为后续任务适应提供了更好的起点。

4.3 Calibration Robustness of GISP

4.3 GISP 的校准鲁棒性

In this section, we study calibration robustness from two perspectives: robustness to calibration budget and robustness to calibration sampling. To be specific, we conduct GISP on Llama2-7B with the same settings in Sec. 4.1 and evaluate on CMQA accuracy.

在本节中，我们从两个角度研究校准鲁棒性：对校准预算的鲁棒性和对校准采样的鲁棒性。具体而言，我们在与第 4.1 节相同的设置下，对 Llama2-7B 进行 GISP，并在 CMQA 准确率上进行评估。

Robustness to calibration budget. We first analyze how varying the calibration budget affects the full pruning trajectory. Starting from the default 512K-token CMQA calibration budget, we subsample the same calibration pool to 1%, 20%, 40%, 60%, and 80%, keep all other settings fixed, rerun the full iterative pruning procedure under each reduced-budget setting, and evaluate 10 spar-

对校准预算的鲁棒性。我们首先分析校准预算变化对完整剪枝轨迹的影响。从默认的 512K-token CMQA 校准预算开始，我们将相同的校准池子采样到 1%、20%、40%、60% 和 80%，保持所有其他设置不变，在每个减少预算的设置下重新运行完整的迭代剪枝过程，并评估 10 spar-

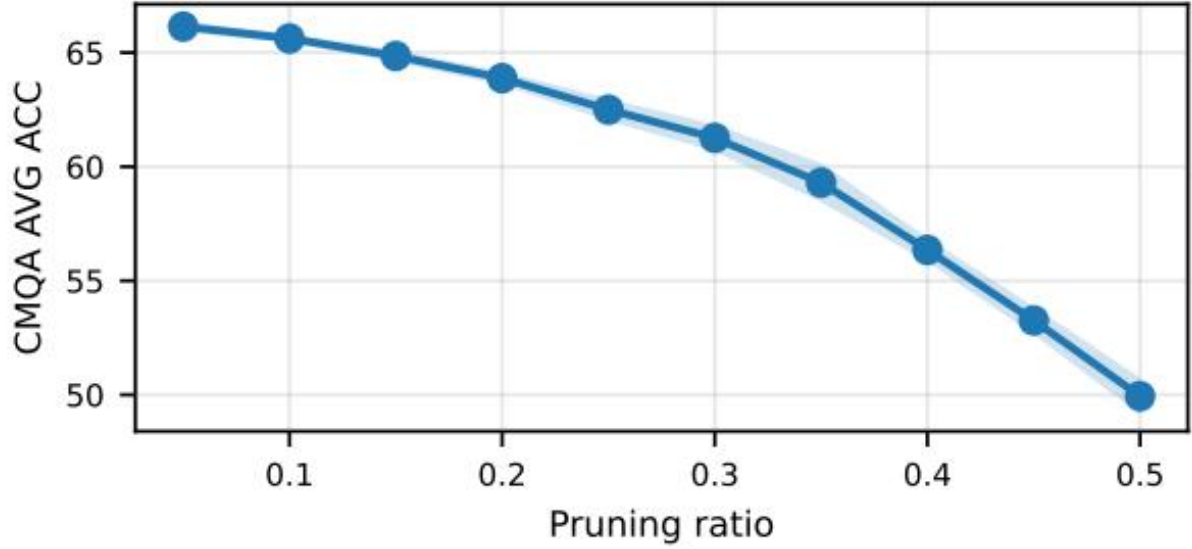


Figure 6: images/8200392e77be54506432cd58e7ab2fc565e9a673319408cad8aca4e0bba5d0ef.jpg

Figure 4: Robustness of the iterative pruning trajectory from GISP under reduced calibration budgets. The shaded region indicates standard deviation.

图 4：在减少校准预算下，GISP 的迭代剪枝轨迹的鲁棒性。阴影区域表示标准差。

sity points from 5% to 50%. Figure 4 summarizes the mean and standard deviation of the resulting pruning trajectories across these reduced-budget settings, while the detailed results can be found in Sec. A.6. As shown in the figure, the overall pruning trajectory remains smooth throughout the full range, and the variation stays small, with the maximum standard deviation below 0.75. This suggests that GISP remains robust even when the available calibration budget is substantially reduced.

大学的积分从 5% 到 50%。图 4 总结了在这些减少预算设置下，所得剪枝轨迹的平均值和标准差，而详细结果可在第 A.6 节中找到。如图所示，整体剪枝轨迹在整个范围内保持平滑，变化保持较小，最大标准差低于 0.75。这表明，即使可用校准预算大幅减少，GISP 仍然保持鲁棒性。

Interestingly, during this analysis, we also observe that even at 1% of the default budget, the resulting CMQA accuracy already remains close to the full-budget setting at the commonly used 20%, 30%, 40%, and 50% pruning ratios. Table 9 provides a direct comparison between the 1% and 100% budgets. The absolute difference is at most 0.66 points, which strongly supports the robustness of GISP under limited calibration data.

有趣的是，在这次分析中，我们还观察到，即使在默认预算的 1% 的情况下，所得到的 CMQA 准确性已经接近于常用的 20%、30%、40% 和 50% 剪枝比率下的全额预算设置。表 9 提供了 1% 和 100% 预算之间的直接比较。绝对差异最大为 0.66 个点，这强烈支持了在有限校准数据下 GISP 的鲁棒性。

Robustness to calibration sampling. To isolate the effect of sample composition from budget size, we additionally resample the calibration subset with random seeds of 0-2, obtaining 63.44 ± 0.07 , 60.50 ± 0.25 , 55.67 ± 0.36 , and 49.11 ± 0.56 average accuracy at 20%, 30%, 40%, and 50% pruning, respectively. These results indicate that GISP is also stable under different calibration samples, with a standard deviation of at most 0.56 points.

对校准样本的鲁棒性。为了将样本组成的影响与预算规模的影响区分开来，我们还使用随机种子 0-2 重新抽样校准子集，分别得到 63.44 ± 0.07 、 60.50 ± 0.25 、 55.67 ± 0.36 和 49.11 ± 0.56 在 20%、30%、40% 和 50% 剪枝下的平均准确率。这些结果表明，GISP 在不同的校准样本下也是稳定的，标准差最多为 0.56 分。

5 Conclusions

5 结论

In this work, we propose GISP, a simple yet effective global iterative structured pruning method for LLMs. GISP prunes globally and iteratively, enabling more flexible, task-aware pruning. It supports once-for-all pruning across multiple sparsity levels and naturally incorporates loss functions tailored to downstream tasks to guide weight importance. Experiments conducted on multiple model families demonstrate clear performance gains compared to prior works, excelling as a task-specific pruner, particularly at high sparsity.

在本工作中，我们提出了 GISP，这是一种简单而有效的面向大语言模型（LLMs）的全局迭代结构化剪枝方法。GISP 全局且迭代地进行剪枝，使剪枝更加灵活，具有任务感知性。它支持在多个稀疏度级别之间进行一次剪枝，自然地结合了针对下游任务量身定制的损失函数，以指导权重的重要性。在多个模型家族上进行的实验表明，与先前的工作相比，性能提升明显，特别是在高稀疏度下，作为任务特定的剪枝器表现出色。

Limitations

限制

One limitation of our method is that, due to its reliance on gradient-based weight importance estimation, it can incur relatively high memory and computational costs. To address this, one could integrate parameter-efficient fine-tuning (PEFT) techniques to accelerate importance computations and reduce the memory footprint—a direction we leave for future work. Additionally, while GISP is designed to be architecture-agnostic and shows promising results on multi-query attention (MQA)-based architectures, we have not yet evaluated it on Mixture-of-Experts (MoE) models due to their significantly larger scale. Extending GISP to MoE architectures remains a valuable direction for future exploration.

我们方法的一个限制是，由于其依赖于基于梯度的权重重要性估计，它可能会导致相对较高的内存和计算成本。为了解决这个问题，可以集成参数高效微调（PEFT）技术，以加速重要性计算并减少内存占用——这是我们未来工作的方向之一。此外，虽然 GISP 被设计为与架构无关，并且在基于多查询注意力（MQA）的架构上表现出有希望的结果，但由于混合专家（MoE）模型的规模显著更大，我们尚未在这些模型上对其进行评估。将 GISP 扩展到 MoE 架构仍然是未来探索的一个有价值的方向。

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